3341 – DISCRETE MATHEMATICS

MODULE 2

Set Theory, Relations and Functions

Solved Previous-Year Questions

Model Questions with Answers

Syllabus covered

- Set representation and types of sets
- Set operations and algebra of sets
- Inclusion–exclusion principle
- Relations: matrix and graph representation
- Types and properties of relations
- Equivalence relations and equivalence classes
- Functions and mappings
- Injective and surjective functions
- Composition and inverse of functions

Exam-oriented notes for students

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1 Basic Concepts and Formulas

1.1 1. Sets

A **set** is a well-defined collection of distinct objects.

Example:

$$A = \{1, 2, 3, 4\}$$

The objects in a set are called its elements.

1.2 2. Number of Subsets

If a set has n elements, the number of possible subsets is

$$\boxed{2^n}$$

The number of proper subsets is

$$\boxed{2^n - 1}$$

The empty set and the set itself are included when counting all subsets.

1.3 3. Cartesian Product

For sets A and B, the Cartesian product is

$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

If

$$|A| = m, \qquad |B| = n,$$

then

$$\boxed{|A \times B| = mn}$$

An ordered pair has an order: generally,

$$(a, b) \neq (b, a).$$

Two ordered pairs are equal if their corresponding components are equal:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

1.4 4. Basic Set Operations

Let A and B be subsets of a universal set U.

Operation	**Meaning**
$A \cup B$	Union: elements belonging to A, B, or both.
$A \cap B$	Intersection: elements common to both A and B.
$A - B$	Difference: elements in A but not in B.
A' or A^c	Complement: elements in U that are not in A.
$\varnothing$	Empty set: a set containing no elements.

1.5 5. Inclusion–Exclusion Principle

For two finite sets:

$$\boxed{n(A \cup B) = n(A) + n(B) - n(A \cap B)}$$

Therefore,

$$\boxed{n(A \cap B) = n(A) + n(B) - n(A \cup B)}$$

For three finite sets:

$$\boxed{\begin{aligned} n(A \cup B \cup C) = {} & n(A) + n(B) + n(C) \\ & - n(A \cap B) - n(A \cap C) \\ & - n(B \cap C) + n(A \cap B \cap C) \end{aligned}}$$

We subtract the pairwise intersections because they were counted twice. Then we add the intersection of all three sets because it was subtracted too many times.

1.6 6. Relations

A relation from A to B is any subset of $A \times B$.

$$\boxed{R \subseteq A \times B}$$

A relation on A is a subset of $A \times A$.

Important properties of a relation R on A:

- **Reflexive:** $(a, a) \in R$ for every $a \in A$.
- **Symmetric:** If $(a, b) \in R$, then $(b, a) \in R$.
- **Antisymmetric:** If $(a, b) \in R$ and $(b, a) \in R$, then $a = b$.
- **Transitive:** If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.

An equivalence relation is a relation that is

$$\boxed{\text{Reflexive + Symmetric + Transitive}}$$

The equivalence class of a is

$$\boxed{[a] = \{x \in A : xRa\}}$$

1.7 7. Functions

A function $f : A \to B$ assigns exactly one output in B to each input in A.

- **Domain:** the set of inputs.
- **Codomain:** the declared set of possible outputs.

- **Range:** the outputs actually obtained.
- **Injective (one-to-one):** different inputs have different outputs.
- **Surjective (onto):** every element of the codomain is an output.
- **Bijective:** both injective and surjective.

For composition:

$$\boxed{(g \circ f)(x) = g(f(x))}$$

Apply f first and then g.

For an invertible function:

$$\boxed{f^{-1}(f(x)) = x, \qquad f(f^{-1}(x)) = x}$$

2 Solved Previous-Year Questions

The following questions are transcribed from the supplied Module 2 previous-year-question PDF. Small wording or notation differences may occur because the original questions were handwritten.

2.1 Part A: One-Mark Questions

Question 1: If A has n elements, find the number of subsets of A.

Concept: Each element has two choices: it can either be included in a subset or excluded from it.

Solution: If there are n elements, the total number of subsets is

$$\boxed{2^n}$$

For example, if $A = \{a, b\}$, its subsets are

$$\varnothing, \{a\}, \{b\}, \{a, b\}.$$

There are $4 = 2^2$ subsets.

Question 2: If $|A| = m$ and $|B| = n$, find $n(A \times B)$.

Concept: Every element of A can be paired with every element of B.

Solution:

$$\boxed{n(A \times B) = mn}$$

For example, if A has 2 elements and B has 3 elements, then $A \times B$ contains $2 \times 3 = 6$ ordered pairs.

Question 3: If $f : A \to B$ is a function, what is A called?

Concept: A function takes its inputs from the domain and maps them into the codomain.

Answer:

$$\boxed{A \text{ is the domain of } f}$$

Question 4: Give an example of a finite set.

Concept: A finite set has a limited number of elements.

Answer:

$$\boxed{A = \{2, 3, 5\}}$$

This set has three elements, so it is finite.

Question 5: If $(x^2 - 1, y + 3) = (3, 4)$, find x and y.

Concept: Two ordered pairs are equal when their first components are equal and their second components are equal.

Solution:

$$x^2 - 1 = 3$$
$$x^2 = 4$$
$$\boxed{x = \pm 2}$$

Also,

$$y + 3 = 4$$
$$\boxed{y = 1}$$

Answer:

$$\boxed{x = 2 \text{ or } x = -2, \qquad y = 1}$$

Question 6: Define an injective function.

Concept: An injective function is also called a one-to-one function.

Answer: A function $f : A \to B$ is injective if

$$\boxed{f(a_1) = f(a_2) \implies a_1 = a_2}$$

for all $a_1, a_2 \in A$.

In simple words, two different inputs cannot have the same output.

2.2 Part B: Three-Mark Questions

Question 1: Find $A \cup B$ and $A \cap B$, if

$$A = \{2, 4, 7, 8\}, \qquad B = \{4, 5, 6, 7\}.$$

Concept: The union contains all distinct elements from both sets. The intersection contains only the common elements.

Solution:

The distinct elements in the two sets are $2, 4, 5, 6, 7, 8$. Therefore,

$$\boxed{A \cup B = \{2, 4, 5, 6, 7, 8\}}$$

The common elements are 4 and 7. Therefore,

$$\boxed{A \cap B = \{4, 7\}}$$

Question 2: Write all subsets of $A = \{1, 2, 3\}$.

Concept: A set containing n elements has 2^n subsets.

Solution: Since A has 3 elements,

$$2^3 = 8.$$

The eight subsets are

$$\boxed{\begin{aligned}\mathcal{P}(A) = \{&\varnothing, \{1\}, \{2\}, \{3\},\\ &\{1, 2\}, \{1, 3\}, \{2, 3\},\\ &\{1, 2, 3\}\}\end{aligned}}$$

Here, $\mathcal{P}(A)$ denotes the power set of A, which is the set containing all subsets of A.

Question 3: Let $f : \mathbb{R} \to \mathbb{R}$, where $f(x) = 3x + 7$. Prove that f is one-to-one.

Concept: To prove injectivity, assume that two inputs have the same output and show that the inputs must be equal.

Solution: Let

$$f(x_1) = f(x_2).$$

Since $f(x) = 3x + 7$,

$$3x_1 + 7 = 3x_2 + 7.$$

Subtracting 7 from both sides gives

$$3x_1 = 3x_2.$$

Dividing by 3,

$$x_1 = x_2.$$

Therefore,

$$\boxed{f \text{ is injective (one-to-one).}}$$

Question 4: If $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are given by $f(x) = \sin x$ and $g(x) = 5x^2$, find $g \circ f$.

Concept: Composition means applying the first function and then the second function:

$$(g \circ f)(x) = g(f(x)).$$

Solution: We are given

$$f(x) = \sin x, \qquad g(x) = 5x^2.$$

Substitute $f(x) = \sin x$ into g:

$$(g \circ f)(x) = g(\sin x).$$

Since $g(t) = 5t^2$,

$$(g \circ f)(x) = 5(\sin x)^2.$$

Hence,

$$\boxed{(g \circ f)(x) = 5\sin^2 x}$$

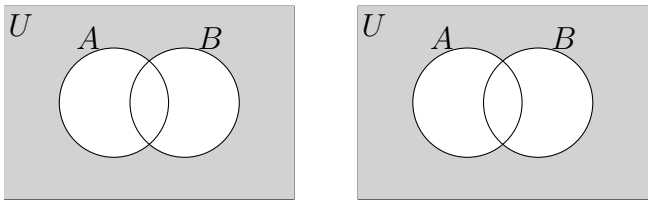
Question 5: Draw appropriate Venn diagrams for $(A \cup B)'$ and $A' \cap B'$.

Concept: A Venn diagram uses a rectangle for the universal set U and circles for the sets.

- $(A \cup B)'$: shade the region outside both circles.
- $A' \cap B'$: shade the region outside A and outside B.

The two expressions represent the same region. This illustrates De Morgan's law:

$$\boxed{(A \cup B)' = A' \cap B'}$$



$(A \cup B)'$ $A' \cap B'$

Question 6: Let $X = \{1, 2, 3, 4\}$ and $R = \{(x, y) : x > y\}$. Draw the graph of R and give its matrix.

Concept: A relation can be represented by ordered pairs, a directed graph, or a matrix.

For the matrix, rows represent the first component x, and columns represent the second component y. Write 1 if xRy, and 0 otherwise.

Solution: Since $x > y$, the valid ordered pairs are

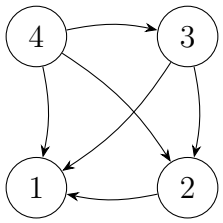
$$\boxed{R = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}}$$

The matrix, using the order $1, 2, 3, 4$ for both rows and columns, is

$$\boxed{M_R = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}}$$

For the directed graph, use vertices $1, 2, 3, 4$ and draw arrows

$$2 \to 1, \quad 3 \to 1, \quad 3 \to 2, \quad 4 \to 1, \quad 4 \to 2, \quad 4 \to 3.$$



Question 7: If $A = \{1, 2, 3, 4\}$, give an example of a relation on A that is reflexive and transitive, but not symmetric.

Concept: A relation is reflexive if it contains every self-pair. It is not symmetric if at least one pair (a, b) is present but the reverse pair (b, a) is absent.

Solution: Take

$$\boxed{R = \{(1,1), (2,2), (3,3), (4,4), (1,2)\}}$$

Reflexive: All required self-pairs $(1,1), (2,2), (3,3), (4,4)$ are present.

Not symmetric: The pair $(1,2)$ is present, but $(2,1)$ is absent.

Transitive: The only non-self pair is $(1,2)$. Combining it with the self-pairs produces no missing pair: for example, $(1,1)$ and $(1,2)$ require $(1,2)$, which is present. Thus the relation is transitive.

Therefore, R is reflexive and transitive, but not symmetric.

2.3 Part C: Seven-Mark Questions

Question 1: Show that $f : \mathbb{N} \to \mathbb{N}$, given by $f(x) = 2x$, is one-to-one and onto.

Concept: A function is injective if equal outputs imply equal inputs. It is onto if every element of the codomain is the output of some input.

Step 1: Prove that f is one-to-one.

Let

$$f(x_1) = f(x_2).$$

Using $f(x) = 2x$,

$$2x_1 = 2x_2.$$

Dividing both sides by 2,

$$x_1 = x_2.$$

Hence f is injective.

Step 2: Test whether f is onto.

For f to be onto, every $y \in \mathbb{N}$ must have a preimage $x \in \mathbb{N}$ satisfying

$$y = 2x.$$

This gives

$$x = \frac{y}{2}.$$

If $y = 3$, then $x = 3/2$, which is not a natural number. Therefore, 3 is not in the range of f.

The range is the set of even natural numbers:

$$\text{Range}(f) = \{2, 4, 6, 8, \ldots\}$$

when $\mathbb{N} = \{1, 2, 3, \ldots\}$.

Thus,

$$\boxed{f \text{ is one-to-one but not onto.}}$$

Exam note: If the intended question is to show that a function is both one-to-one and onto, the function or its codomain may have been written incorrectly in the source.

Question 2: Explain different types of relations in sets with examples.

Concept: A relation is a subset of a Cartesian product. Relations can be classified according to the ordered pairs they contain.

1. Inverse relation

If R is a relation from A to B, its inverse relation is

$$\boxed{R^{-1} = \{(y, x) : (x, y) \in R\}}.$$

Example:

$$R = \{(1, 2), (3, 2), (5, 1)\}.$$

Then

$$\boxed{R^{-1} = \{(2, 1), (2, 3), (1, 5)\}}.$$

The inverse reverses every ordered pair.

2. Universal relation

A relation on A is universal if it contains every ordered pair in $A \times A$:

$$\boxed{R = A \times A}.$$

Example: If $A = \{1, 2\}$, then

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2)\}.$$

3. Empty relation

A relation is empty if it contains no ordered pairs:

$$\boxed{R = \varnothing}.$$

Example: Let $A = \{3, 4\}$ and define

$$R = \{(x, y) : x + y > 10\}.$$

No pair from $A \times A$ satisfies this condition, so

$$R = \varnothing.$$

4. Identity relation

The identity relation on A contains exactly the self-pairs:

$$\boxed{I_A = \{(x, x) : x \in A\}}.$$

Example: If $A = \{1, 2\}$, then

$$I_A = \{(1, 1), (2, 2)\}.$$

5. Reflexive relation

A relation is reflexive if

$$\boxed{(x, x) \in R \text{ for every } x \in A}.$$

Example: For $A = \{1, 2\}$,

$$R = \{(1, 1), (1, 2), (2, 2)\}$$

is reflexive because both required self-pairs are present.

6. Symmetric relation

A relation is symmetric if

$$\boxed{(x, y) \in R \implies (y, x) \in R}.$$

Example:

$$R = \{(1, 1), (2, 2), (1, 2), (2, 1)\}$$

is symmetric because each pair has its reverse.

7. Transitive relation

A relation is transitive if

$$\boxed{(x, y) \in R \text{ and } (y, z) \in R \implies (x, z) \in R}.$$

Example:

$$R = \{(1, 1), (1, 2), (2, 2)\}$$

is transitive. In particular, the chain $(1, 2)$ and $(2, 2)$ requires $(1, 2)$, which is present.

Conclusion: When a question asks for types of relations, write the definition and a suitable example for each type requested.

Question 3: If $A \cup B$ has 50 elements, A has 28 elements and B has 32 elements, find $n(A \cap B)$.

Concept: When the sizes of two sets and their union are known, use the two-set inclusion–exclusion formula.

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Given:

$$n(A) = 28, \quad n(B) = 32, \quad n(A \cup B) = 50.$$

Substituting:

$$50 = 28 + 32 - n(A \cap B)$$

$$50 = 60 - n(A \cap B)$$

Therefore,

$$n(A \cap B) = 60 - 50.$$

$$\boxed{n(A \cap B) = 10}$$

There are 10 elements common to both sets.

Question 4: In a group, 49 study Physics, 37 study English and 21 study Biology. Nine study Physics and English, five study English and Biology, four study Physics and Biology, and three study all three subjects. Find the number of students in the group.

Concept: For three sets, add the sizes of the individual sets, subtract the pairwise intersections, and add the intersection of all three.

Let P, E and B represent the students studying Physics, English and Biology, respectively.

Given:

$$\begin{aligned} n(P) = 49, \qquad & n(E) = 37, \quad n(B) = 21, \\ n(P \cap E) = 9, \qquad & n(E \cap B) = 5, \\ n(P \cap B) = 4, \quad & n(P \cap E \cap B) = 3. \end{aligned}$$

Using inclusion–exclusion:

$$\begin{aligned} n(P \cup E \cup B) = {} & n(P) + n(E) + n(B) \\ & - n(P \cap E) - n(E \cap B) \\ & - n(P \cap B) + n(P \cap E \cap B). \end{aligned}$$

Substitute the values:

$$\begin{aligned} n(P \cup E \cup B) &= 49 + 37 + 21 \\ &\quad - 9 - 5 - 4 + 3 \\ &= 107 - 18 + 3 \\ &= 92. \end{aligned}$$

$$\boxed{n(P \cup E \cup B) = 92}$$

Assuming every student in the group studies at least one of the three subjects, the group contains 92 students.

Question 5: Find the inverse of $f(x) = 6x + 3$. Verify it by computing the composition.

Concept: To find an inverse, write $y = f(x)$, interchange x and y, and solve for y.

Step 1: Find the inverse.

Given:

$$y = 6x + 3.$$

Subtract 3:

$$y - 3 = 6x.$$

Divide by 6:

$$x = \frac{y-3}{6}.$$

Interchanging x and y, we obtain

$$\boxed{f^{-1}(x) = \frac{x-3}{6}}.$$

Step 2: Verify $f(f^{-1}(x)) = x$.

$$\begin{aligned} f(f^{-1}(x)) &= f\left(\frac{x-3}{6}\right) \\ &= 6\left(\frac{x-3}{6}\right) + 3 \\ &= x - 3 + 3 \\ &= x. \end{aligned}$$

Step 3: Verify $f^{-1}(f(x)) = x$.

$$\begin{aligned} f^{-1}(f(x)) &= f^{-1}(6x+3) \\ &= \frac{(6x+3)-3}{6} \\ &= \frac{6x}{6} \\ &= x. \end{aligned}$$

Both compositions give x. Therefore,

$$\boxed{f^{-1}(x) = \frac{x-3}{6}}$$

is the inverse of $f(x) = 6x + 3$.

3 Additional Explanations for the Syllabus

3.1 Algebra of Sets

The following laws are useful for simplifying set expressions.

Law	**Identity**
Commutative	$A \cup B = B \cup A$ $A \cap B = B \cap A$

Associative	$(A \cup B) \cup C = A \cup (B \cup C)$ $(A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
Identity	$A \cup \varnothing = A$ $A \cap U = A$
Domination	$A \cup U = U$ $A \cap \varnothing = \varnothing$
Idempotent	$A \cup A = A$ $A \cap A = A$
Complement	$A \cup A' = U$ $A \cap A' = \varnothing$
Double complement	$(A')' = A$
De Morgan's laws	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$

3.2 Equivalence Relations and Equivalence Classes

A relation R on a set A is an equivalence relation only if it satisfies all three properties:

1. Reflexive
2. Symmetric
3. Transitive

Example

Let

$$A = \{1, 2, 3, 4\}$$

and

$$R = \{(1,1), (1,3), (3,1), (3,3), (2,2), (2,4), (4,2), (4,4)\}.$$

Each element is related to itself, so R is reflexive.

Every non-self pair has its reverse, so R is symmetric.

The elements are grouped into $\{1, 3\}$ and $\{2, 4\}$, and all required pairs within each group are present. Therefore, R is transitive.

Hence R is an equivalence relation, and its equivalence classes are

$$\boxed{[1] = [3] = \{1, 3\}}$$

and

$$\boxed{[2] = [4] = \{2, 4\}}.$$

3.3 Domain, Codomain and Range

Consider

$$f : A \to B, \qquad A = \{1, 2, 3\}, \quad B = \{1, 4, 9, 16\},$$

where $f(x) = x^2$.

The outputs are

$$f(1) = 1, \quad f(2) = 4, \quad f(3) = 9.$$

Therefore,

$$\text{Domain} = \{1, 2, 3\},$$
$$\text{Codomain} = \{1, 4, 9, 16\},$$
$$\text{Range} = \{1, 4, 9\}.$$

The range need not equal the codomain. This function is not onto because 16 is in the codomain but is not an output.

3.4 Injective and Surjective Functions

Injective: If $f(a) = f(b)$ implies $a = b$, the function is one-to-one.

Surjective: If every element of the codomain has at least one preimage in the domain, the function is onto.

Bijective: A function is bijective if it is both injective and surjective. A bijective function has an inverse function from its codomain to its domain.

3.5 Composition of Functions

Suppose

$$f(x) = x + 2, \qquad g(x) = 3x.$$

Then

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(x + 2) \\ &= 3(x + 2) \\ &= \boxed{3x + 6}.\end{aligned}$$

The order matters. For the reverse composition,

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(3x) \\ &= \boxed{3x + 2}.\end{aligned}$$

Thus, in general, $g \circ f \neq f \circ g$.

4 Model Questions Based on the Previous-Year Pattern

The following questions are newly prepared for practice. They follow the topics and styles of the supplied previous-year questions. They are not presented as actual past examination questions.

4.1 Part A: One-Mark Questions

Question 1. If A has 5 elements, how many subsets does it have?

Answer:

$$2^5 = \boxed{32}$$

Explanation: A set with n elements has 2^n subsets.

Question 2. If $|A| = 4$ and $|B| = 3$, find $|A \times B|$.

Answer:

$$|A \times B| = 4 \times 3 = \boxed{12}$$

Explanation: Each of the four elements of A can be paired with each of the three elements of B.

Question 3. Define a surjective function.

Answer: A function $f : A \to B$ is surjective if every element of B is the image of at least one element of A.

Explanation: For an onto function, range equals codomain.

Question 4. Give an example of an identity relation on $A = \{a, b\}$.

Answer:

$$\boxed{I_A = \{(a, a), (b, b)\}}$$

Explanation: The identity relation contains only self-pairs.

Question 5. If $f(x) = 4x - 1$, find $f(2)$.

Answer:

$$f(2) = 4(2) - 1 = 8 - 1 = \boxed{7}$$

Explanation: Substitute $x = 2$ into the given function.

4.2 Part B: Three-Mark Questions

Question 1. Let

$$A = \{1, 2, 3, 5\}, \qquad B = \{2, 4, 5, 6\}.$$

Find $A \cup B$, $A \cap B$ and $A - B$.

Answer:

$$\boxed{A \cup B = \{1, 2, 3, 4, 5, 6\}}$$

$$\boxed{A \cap B = \{2, 5\}}$$

$$\boxed{A - B = \{1, 3\}}$$

Explanation: The union combines all distinct elements, the intersection contains common elements, and $A - B$ contains elements of A that are not in B.

Question 2. Write all subsets of $A = \{a, b\}$.

Answer:

$$\boxed{\mathcal{P}(A) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}}$$

Explanation: There are $2^2 = 4$ subsets.

Question 3. Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 5x - 2$. Prove that f is injective.

Answer: Assume $f(x_1) = f(x_2)$.

$$\begin{aligned} 5x_1 - 2 &= 5x_2 - 2 \\ 5x_1 &= 5x_2 \\ x_1 &= x_2. \end{aligned}$$

Therefore, $\boxed{f \text{ is injective}}$.

Explanation: Equal outputs imply equal inputs.

Question 4. If $f(x) = x + 1$ and $g(x) = 2x$, find $(g \circ f)(x)$.

Answer:

$$\begin{aligned} (g \circ f)(x) &= g(f(x)) \\ &= g(x + 1) \\ &= 2(x + 1) \\ &= \boxed{2x + 2}. \end{aligned}$$

Explanation: Apply f first, then apply g.

Question 5. Let $A = \{1, 2, 3\}$ and

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2)\}.$$

Is R reflexive? Is it symmetric?

Answer: The relation is reflexive because all three self-pairs are present. It is not symmetric because $(1, 2) \in R$, but $(2, 1) \notin R$.

Explanation: Check all self-pairs for reflexivity and check whether each ordered pair has its reverse for symmetry.

4.3 Part C: Seven-Mark Questions

Question 1. In a class, 36 students study Mathematics, 28 study Computer Science, and 50 study at least one of the two subjects. Find the number who study both subjects.

Solution:

Let M be the set of students studying Mathematics and C the set studying Computer Science.

Given:

$$n(M) = 36, \quad n(C) = 28, \quad n(M \cup C) = 50.$$

Using inclusion–exclusion:

$$n(M \cup C) = n(M) + n(C) - n(M \cap C).$$

Substitute:

$$50 = 36 + 28 - n(M \cap C).$$

$$50 = 64 - n(M \cap C).$$

Therefore,

$$n(M \cap C) = 64 - 50.$$

$$\boxed{n(M \cap C) = 14}$$

Explanation: Students studying both subjects are counted in both individual totals, so the intersection is subtracted once.

Question 2. In a group, 40 students study Mathematics, 32 study Physics and 25 study Chemistry. Also,

$$n(M \cap P) = 10, \quad n(P \cap C) = 8, \quad n(M \cap C) = 6$$

and

$$n(M \cap P \cap C) = 4.$$

Find the number of students studying at least one subject.

Solution:

Using the three-set inclusion–exclusion formula:

$$\begin{aligned} n(M \cup P \cup C) = {} & n(M) + n(P) + n(C) \\ & - n(M \cap P) - n(P \cap C) \\ & - n(M \cap C) + n(M \cap P \cap C). \end{aligned}$$

Substitute:

$$\begin{aligned} n(M \cup P \cup C) &= 40 + 32 + 25 \\ &\quad - 10 - 8 - 6 + 4 \\ &= 97 - 24 + 4 \\ &= \boxed{77}. \end{aligned}$$

Explanation: Add the three set sizes, subtract each pairwise intersection, and add the common intersection of all three.

Question 3. Let $A = \{1, 2, 3\}$ and

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}.$$

Determine whether R is an equivalence relation. If it is, find the equivalence classes.

Solution:

Reflexive: All self-pairs $(1, 1), (2, 2), (3, 3)$ are present. Thus, R is reflexive.

Symmetric: The pair $(1, 2)$ is present and its reverse $(2, 1)$ is also present. The self-pairs are automatically symmetric. Thus, R is symmetric.

Transitive: The elements 1 and 2 are related to each other, and the required pairs within $\{1, 2\}$ are present. The element 3 is related only to itself. Hence all required transitive pairs are present.

Therefore, R is an equivalence relation.

Its equivalence classes are

$$\boxed{[1] = [2] = \{1, 2\}}$$

and

$$\boxed{[3] = \{3\}}.$$

Explanation: An equivalence relation divides a set into non-overlapping groups of related elements.

Question 4. Find the inverse of $f(x) = 4x + 7$ and verify the answer by composition.

Solution:

Let

$$y = 4x + 7.$$

Then

$$y - 7 = 4x$$

and

$$x = \frac{y - 7}{4}.$$

Interchange x and y:

$$\boxed{f^{-1}(x) = \frac{x - 7}{4}}.$$

Verify the first composition:

$$\begin{aligned} f(f^{-1}(x)) &= 4\left(\frac{x - 7}{4}\right) + 7 \\ &= x - 7 + 7 \\ &= x. \end{aligned}$$

Verify the second composition:

$$\begin{aligned} f^{-1}(f(x)) &= \frac{(4x + 7) - 7}{4} \\ &= \frac{4x}{4} \\ &= x. \end{aligned}$$

Thus,

$$\boxed{f^{-1}(x) = \frac{x - 7}{4}}.$$

Explanation: A function and its inverse undo each other's operations, so both compositions give the original input.

Question 5. Let $X = \{1, 2, 3\}$ and

$$R = \{(x, y) \in X \times X : x \leq y\}.$$

Write R and its matrix.

Solution:

The Cartesian product contains all ordered pairs from X. Retain only those satisfying $x \leq y$.

Thus,

$$\boxed{R = \{(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\}}$$

For the matrix, rows represent x and columns represent y. Write 1 when $x \leq y$, and 0 otherwise.

$$\boxed{M_R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}$$

Explanation: The first row contains three ones because $1 \leq 1, 2, 3$. The second row has ones for columns 2 and 3, and the third row has a one only for column 3.

5 Quick Revision Table

Topic	**Remember**
Number of subsets	2^n
Cartesian product size	$\lvert A \times B \rvert = \lvert A \rvert \lvert B \rvert$
Union of two sets	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Union of three sets	Add individual sizes, subtract pairwise intersections, then add the triple intersection.
Relation	$R \subseteq A \times B$
Reflexive	Every self-pair is present.
Symmetric	Every pair has its reverse.
Transitive	If aRb and bRc, then aRc.
Equivalence relation	Reflexive, symmetric and transitive.
Injective function	$f(a) = f(b) \implies a = b$.
Surjective function	Range equals codomain.
Composition	$(g \circ f)(x) = g(f(x))$.
Inverse function	$f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$.

6 Exam Tips

1. For a one-mark question, write the definition or formula directly and clearly.
2. For set operations, list each element only once in the union.
3. For inclusion–exclusion questions, write the formula before substituting the values.
4. For a three-set problem, do not forget to add the intersection of all three sets at the end.
5. To test a relation, check each required property separately.

6. For a relation matrix, clearly state the order of the rows and columns.
7. To prove injectivity, begin with $f(x_1) = f(x_2)$.
8. To test surjectivity, check whether every codomain element has a preimage.
9. For composition, apply the function inside the brackets first.
10. For an inverse, interchange x and y after solving for the original input.
11. When verifying an inverse, show both compositions if the question asks for verification.

End of Module 2 Notes

Discrete Mathematics – Set Theory, Relations and Functions